

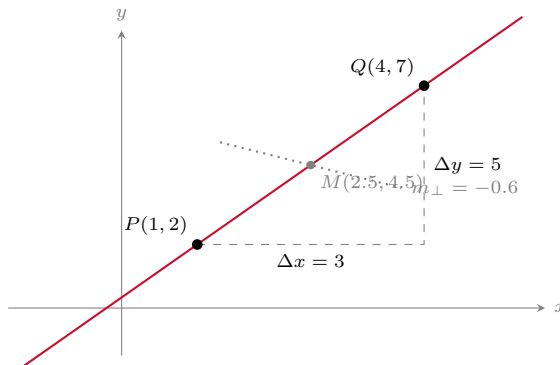
Lines: the offside line and the touchline

THE ONE IDEA

A line is a gradient and one point. How steep, and one place it has been, and you know everywhere it goes.

Board order: two points give a gradient, a distance and a midpoint; then $y = mx + c$; parallel and perpendicular; the general form $Ax + By = C$; point and gradient; two points.

THE PICTURE: EXAMPLE 7



$$m = \frac{\Delta y}{\Delta x} = 1.67$$

$$\text{Gradient } m = \frac{7 - 2}{4 - 1} = \frac{5}{3}$$

$$\text{Point and gradient: } y - 2 = \frac{5}{3}(x - 1).$$

$$\text{Times 3: } 3y - 6 = 5x - 5, \text{ so } -5x + 3y = 1.$$

$$\text{Midpoint } M(2.5, 4.5); \text{ perpendicular gradient } -\frac{3}{5}.$$

$m > 0$: increasing $m < 0$: decreasing $m = 0$: horizontal, $y = c$ vertical: gradient not defined, $x = c$

THREE WAYS TO WRITE A LINE

Form	Formula	What it hands you	In the example
Gradient-intercept	$y = mx + c$	Gradient m (rise per unit of run, $m = \tan \theta$) and the y -intercept c . Horizontal: $y = c$. Vertical: $x = c$, which is not a function, so not a case of this form.	$y = 3x - 1$ climbs 3 per step and crosses the y -axis at -1 .
Point-gradient	$y - y_0 = m(x - x_0)$	The line through a known point (x_0, y_0) with gradient m . With two points, find m first, then use either point.	Through $(1, 2)$ with $m = 3$: $y - 2 = 3(x - 1)$, so $y = 3x - 1$.
General	$Ax + By = C$	Integer coefficients, no fractions. $B = 0$ is vertical, $A = 0$ is horizontal. Clear denominators by multiplying every term.	$y = \frac{1}{2}x + \frac{2}{3}$ times 6: $-3x + 6y = 4$.

TWO POINTS

Gradient $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$. Distance $d = \sqrt{(\Delta x)^2 + (\Delta y)^2}$. Midpoint: average the x 's, average the y 's. Example 1: $A(1, 4)$, $B(7, 12)$: $m = \frac{8}{6} = \frac{4}{3}$, $d = 10$, $M(4, 8)$.

PARALLEL AND PERPENDICULAR

Parallel: same gradient. Perpendicular: $m_2 = -\frac{1}{m_1}$, flip and change sign, so the product is -1 . $y = 3x + 5$ and $y = -\frac{1}{3}x + 8$ are perpendicular.

IN PLAIN ENGLISH

The gradient is how much you climb for each step to the right. Give me a gradient and one point and the line is fixed; a second point only tells you the gradient.

TRAPS

- Δy over Δx , rise over run, never the other way round.
- A vertical line has no gradient. Not gradient zero: no gradient. Its equation is $x = c$.
- Perpendicular means flip and change sign. Changing only the sign gives a reflection, not a right angle.

WHEN LINES COME BACK (IN 2.3, AND IN EVERY TANGENT YOU WILL EVER DRAW)

1. A gradient is asked for: Δy over Δx from two points, or read m off $y = mx + c$.
2. An equation of a line is asked for: $y - y_0 = m(x - x_0)$ first, then tidy into the form the question names.
3. The word perpendicular appears: write $-1/m$ before anything else.

Angle of a line: $m = \tan \theta$, where θ is the angle the line makes with the x -axis.

Three checks and one question

This lesson has already happened. Do the three checks cold, without the notes; each should take under two minutes. The question is for the next time lines appear in class.

1. [check 1]

$A(2, -1)$ and $B(8, 7)$. Find the gradient, the distance and the midpoint.

What should come out: $m = \frac{8}{6} = \frac{4}{3}$, $d = \sqrt{36 + 64} = 10$, $M(5, 3)$.

What came out:

2. [check 2]

The line through $(1, 2)$ and $(4, 7)$, in the form $ax + by = c$ with integers.

What should come out: $m = \frac{5}{3}$; $y - 2 = \frac{5}{3}(x - 1)$; $3y - 6 = 5x - 5$; $-5x + 3y = 1$ (or $5x - 3y = -1$).

What came out:

3. [check 3]

The line through $(4, 7)$ perpendicular to $y = 3x + 5$, in the form $y = mx + c$.

What should come out: $m = -\frac{1}{3}$; $y - 7 = -\frac{1}{3}(x - 4)$; $y = -\frac{1}{3}x + \frac{25}{3}$.

What came out:

4. [the question]

A vertical line is not a function, so $x = c$ is not a case of $y = mx + c$. Is that also why $Ax + By = C$ is the more general form, because it is the only one that can write a vertical line?

What should come out: Yes. With $B = 0$ the general form gives $x = C/A$; no version of $y = mx + c$ can. Ask it when the general form next appears.

What came out:
